

Question 1

3 marks 101

Determine and simplify the 8th term in the binomial expansion of $\left(x - \frac{2}{x^3}\right)^{10}$.

Question 2

2 marks

102

The temperature of hot chocolate is recorded as it cools down. The data shows that the temperature cools according to the equation:

$$T = 82(0.87)^t + 16$$

where T is the temperature of the hot chocolate, in degrees Celsius after t minutes and

t is the time, in minutes, after the hot chocolate is made.

In order to avoid burns, hot chocolate should not be served at a temperature higher than 71°C . Determine, algebraically, the amount of time it will take the hot chocolate to reach this temperature.

Question 3

3 marks 103

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$3\sin^2\theta + 6\sin\theta + 2 = 0$$

Question 4

3 marks 104

A committee of 4 people is to be selected from 6 high school students and 5 middle years students.

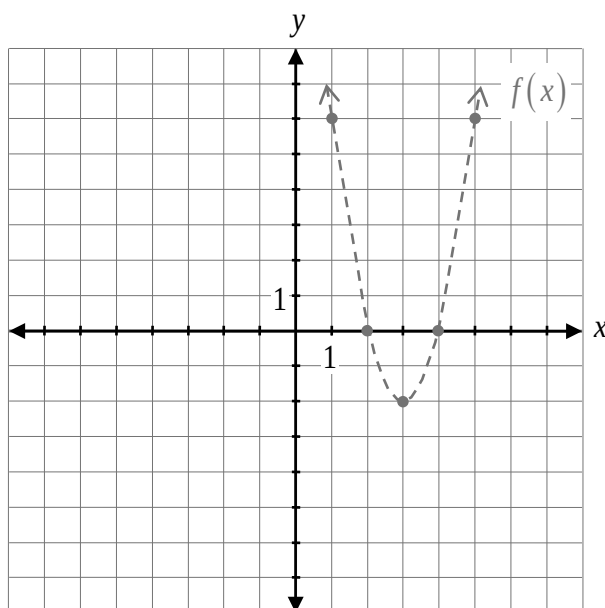
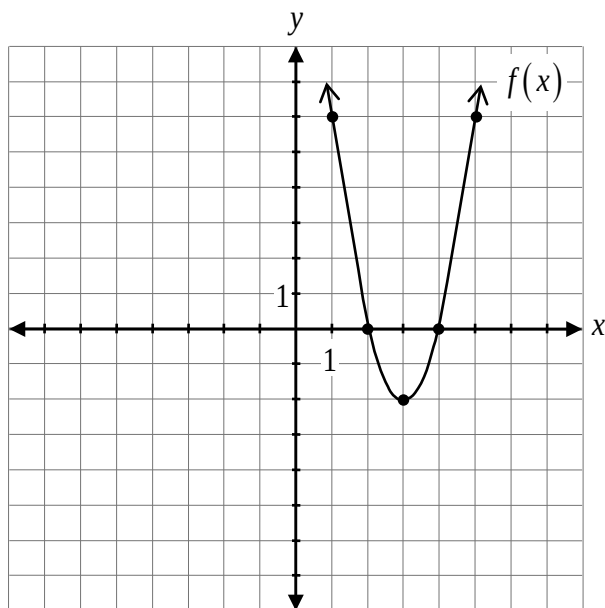
Determine the number of possible committees if the committee must have at least 3 high school students.

Note: A calculator is not required for the remaining test questions.

Question 5

2 marks 105

Given the graph of $y = f(x)$, sketch the graph of $y = -|f(x)|$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 6

3 marks 106

Describe the transformations used to obtain the graph of the function $y = 4f\left(\frac{x}{2}\right) - 3$ from the graph of $y = f(x)$.

Question 7

2 marks 107

If $\log 4 = m$ and $\log 3 = n$, express $\log 48$ in terms of m and n .

Question 8

2 marks 108

State the domain and range of the radical function, $f(x) = -3\sqrt{x-8} + 1$.

Domain: _____

Range: _____

Question 9

1 mark

109

Explain why the graph of $y = \tan x$ does not have an amplitude.

Question 10

3 marks 110

Prove the following identity for all permissible values of θ .

$$\frac{\sec \theta - \sin^2 \theta \sec \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Left-Hand Side

Right-Hand Side

Question 11**2 marks** 111

Given that $x = 3$ is one of the zeros of $p(x) = x^3 - 7x - 6$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 12

3 marks 112

Solve, algebraically.

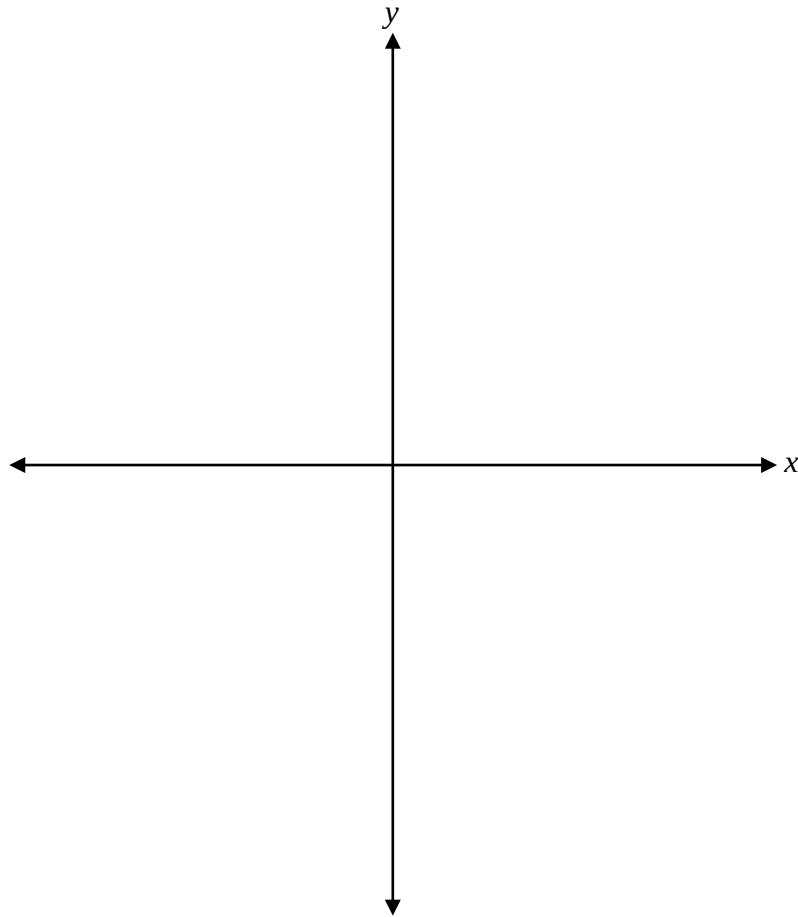
$${}_{n+1}P_2 = 6$$

Question 13

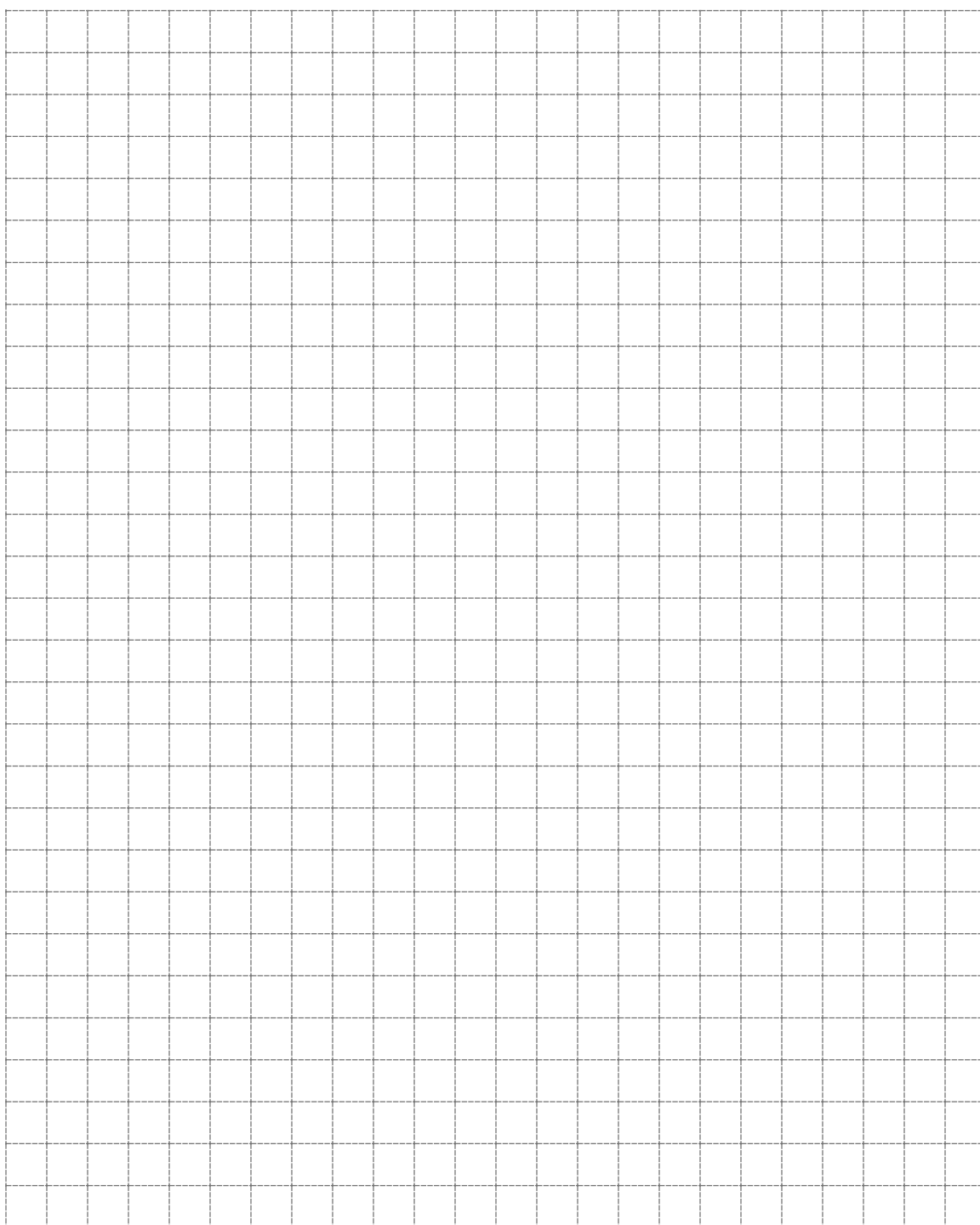
2 marks 113

Express $\log(3x - 1) - \log x + \log 9$ as a single logarithm.

Sketch the graph of $P(x) = -(x-2)^2(x+3)$.



No marks will be awarded for work done on this page.



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